

Fundamentals of Linear Control:

A Concise Approach

Maurício C. de Oliveira

Chapter 5: State-Space Models and Linearization

linearcontrol.info/fundamentals

5.1 Realization of Dynamic Systems

Integrators

Water tank

$$y(t) = \frac{1}{A} \int_0^t u(\tau) d\tau$$

y: water level, u: input flow

A: cross-section area

Fly-wheel without friction

$$\omega(t) = \frac{1}{J} \int_0^t f(\tau) d\tau$$

ω : angular speed, f : input torque

J: moment of inertia

Capacitor

$$v(t) = \frac{1}{C} \int_0^t i(\tau) d\tau$$

v: voltage, i: input current

C: capacitance

Trapezoidal integration rule

$$\begin{aligned} y(kT) - y(kT - T) &= \int_{kT-T}^{kT} u(t) dt \\ &\approx \frac{T}{2} (u(kT) + u(kT - T)) \end{aligned}$$

y: integrated output, u: input

T: sampling period

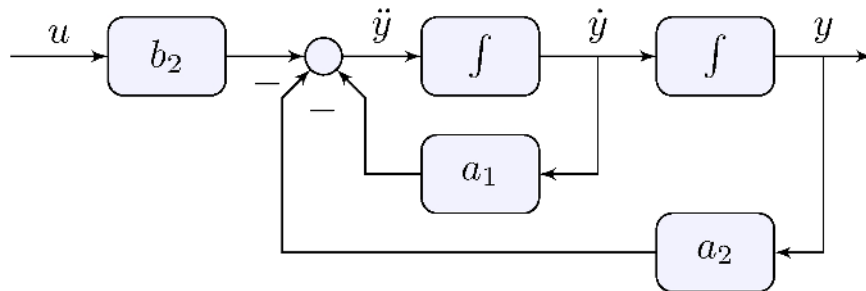
Second-order model

$$\ddot{y} + a_1 \dot{y} + a_2 y = b_2 u$$

Isolate highest derivative

$$\ddot{y} = b_2 u - a_1 \dot{y} - a_2 y$$

Block-diagram



General second-order model

$$\ddot{y} + a_1 \dot{y} + a_2 y = b_0 \ddot{u} + b_1 \dot{u} + b_2 u$$

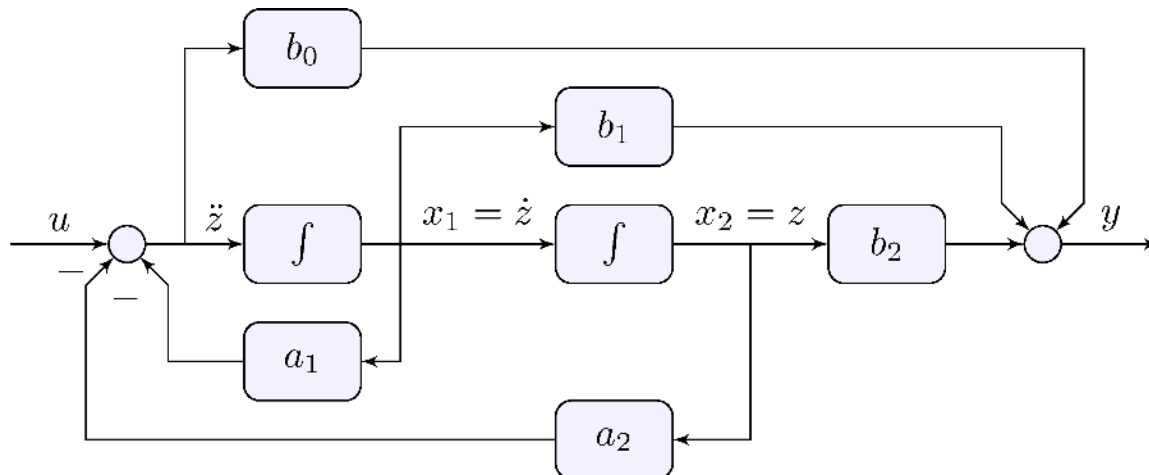
Linearity

$$\ddot{y} + a_1 \dot{y} + a_2 y = u_0 + u_1 + u_2, \quad u_0 = b_0 \ddot{u}, \quad u_1 = b_1 \dot{u}, \quad u_2 = b_2 u$$

can be broken down into

$$\ddot{z} + a_1 \dot{z} + a_2 z = u, \quad y = b_0 \ddot{z} + b_1 \dot{z} + b_2 z$$

Block-diagram



General second-order model

$$\ddot{y} + a_1 \dot{y} + a_2 y = b_0 \ddot{u} + b_1 \dot{u} + b_2 u$$

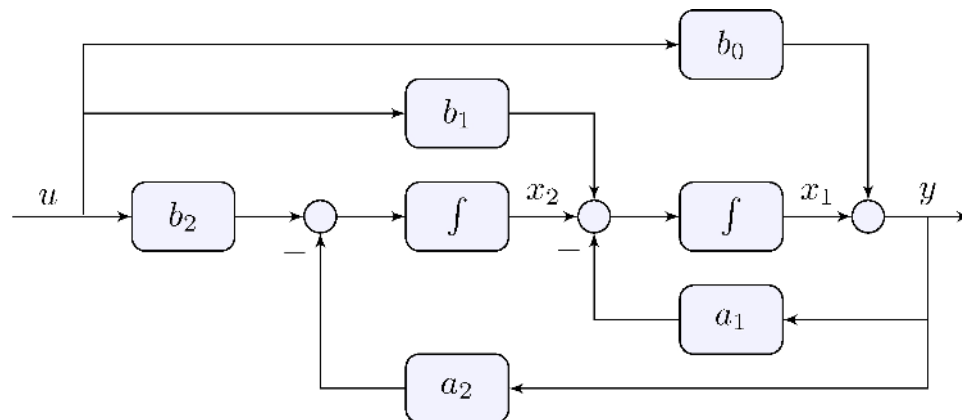
Isolate highest derivative

$$\ddot{y} = b_0 \ddot{u} + (b_1 \dot{u} - a_1 \dot{y}) + (b_2 u - a_2 y)$$

Integrate twice

$$y(t) = b_0 u(t) + \int_0^t \left[b_1 u(\tau) - a_1 y(\tau) + \int_0^\tau (b_2 u(\sigma) - a_2 y(\sigma)) d\sigma \right] d\tau$$

Block-diagram



Generalizes to higher-order models

$$y^{(n)}(t) + a_1 y^{(n-1)}(t) + \cdots + a_n y(t) = b_0 u^{(m)}(t) + b_1 y^{(m-1)}(t) + \cdots + b_m u(t)$$

Must be proper: $m \leq n$

If not proper, needs differentiator

$$y = \dot{u}$$

Impossible to realize!

Differentiator

$$u(t) = \cos(\omega t) \quad \Longrightarrow \quad y_{ss}(t) = -|\omega| \sin(\omega t)$$

Unbounded amplification at high frequencies!

Integrators

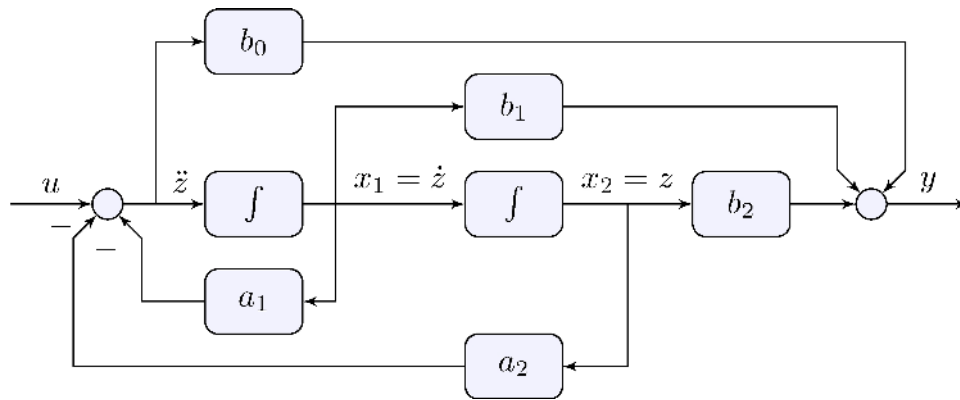
$$u(t) = \cos(\omega t) \quad \Longrightarrow \quad y_{ss}(t) = \frac{1}{|\omega|} \sin(\omega t)$$

Unbounded storage at low frequencies: can be dealt with in closed-loop

See book for more

5.2 State-Space Models

Controllable Realization



Integrator output = state

$$x_1 = \dot{z}, \quad x_2 = z$$

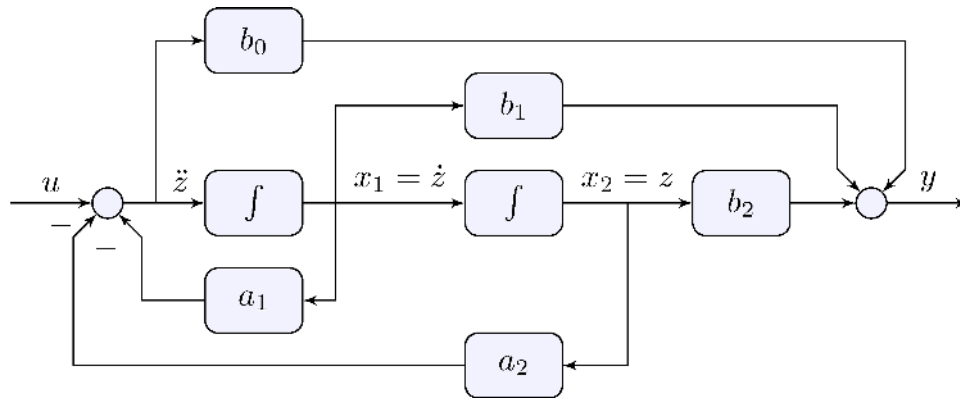
Integrator input = equations

$$\dot{x}_1 = u - a_1 x_1 - a_2 x_2, \quad \dot{x}_2 = x_1$$

Output

$$y = b_0 \dot{x}_1 + b_1 x_1 + b_2 x_2$$

Controllable Realization



Integrator output = state

$$x_1 = \dot{z}, \quad x_2 = z$$

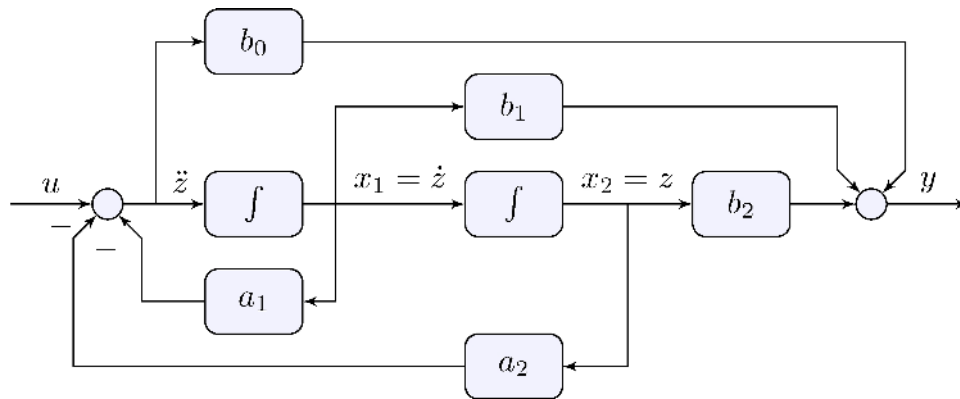
Integrator input = equations

$$\dot{x}_1 = u - a_1 x_1 - a_2 x_2, \quad \dot{x}_2 = x_1$$

Output

$$y = (b_1 - a_1 b_0) x_1 + (b_2 - a_2 b_0) x_2 + b_0 u$$

Controllable Realization



All equations

$$\dot{x}_1 = u - a_1 x_1 - a_2 x_2$$

$$\dot{x}_2 = x_1$$

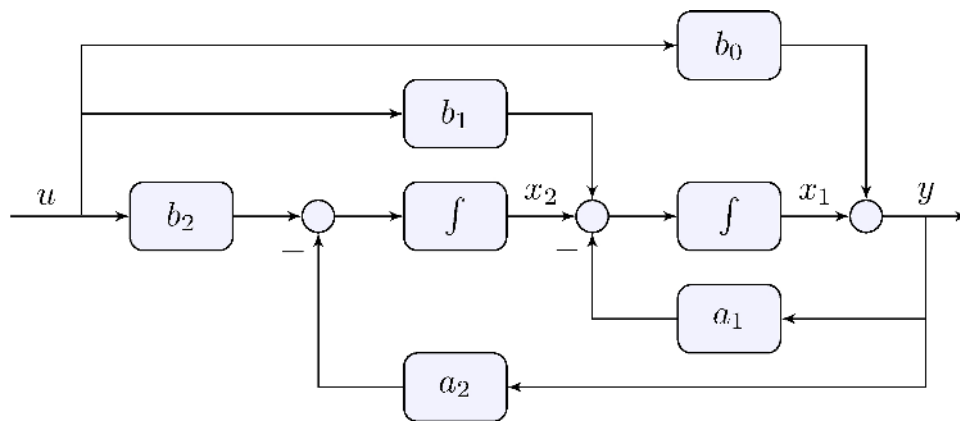
$$y = (b_1 - a_1 b_0) x_1 + (b_2 - a_2 b_0) x_2 + b_0 u$$

State-space form

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{bmatrix} -a_1 & -a_2 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

$$y = [b_1 - a_1 b_0 \quad b_2 - a_2 b_0] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + [b_0] u$$

Observable Realization



Integrator output = state

$$x_1, \quad x_2$$

Integrator input = equations

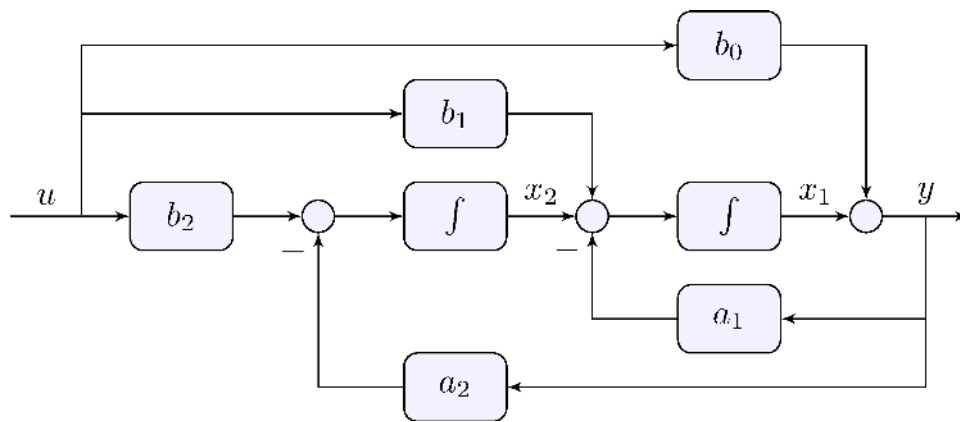
$$\dot{x}_1 = b_1 u - a_1 y + x_2$$

$$\dot{x}_2 = b_2 u - a_2 y$$

Output

$$y = b_0 u + x_1$$

Observable Realization



Integrator output = state

$$x_1, \quad x_2$$

Integrator input = equations

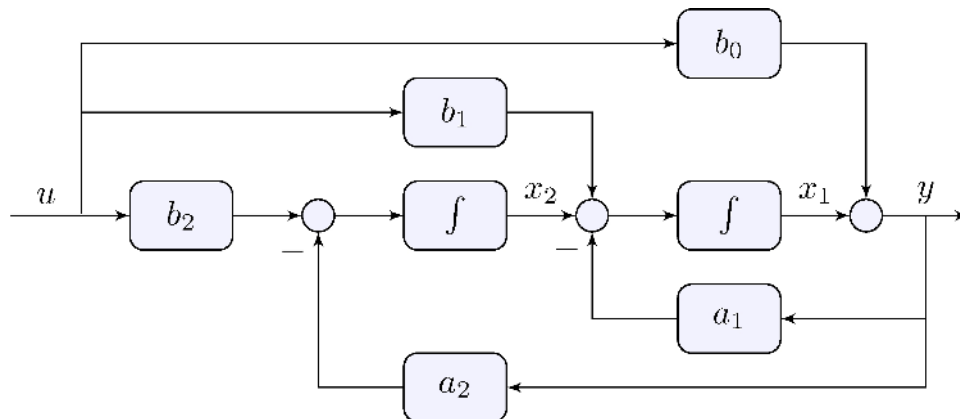
$$\dot{x}_1 = -a_1x_1 + x_2 + (b_1 - a_1b_0)u$$

$$\dot{x}_2 = -a_2x_1 + (b_2 - a_2b_0)u$$

Output

$$y = b_0u + x_1$$

Observable Realization



All equations

$$\dot{x}_1 = -a_1 x_1 + x_2 + (b_1 - a_1 b_0)u$$

$$\dot{x}_2 = -a_2 x_1 + (b_2 - a_2 b_0)u$$

$$y = b_0 u + x_1$$

State-space form

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{bmatrix} -a_1 & 1 \\ -a_2 & 0 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{bmatrix} b_1 - a_1 b_0 \\ b_2 - a_2 b_0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + [b_0]u$$

Second-order model

$$\ddot{y} + a_1 \dot{y} + a_2 y = b_0 \ddot{u} + b_1 \dot{u} + b_2 u$$

State-space form I

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{bmatrix} -a_1 & -a_2 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$
$$y = [b_1 - a_1 b_0 \quad b_2 - a_2 b_0] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + [b_0] u$$

State-space form II

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{bmatrix} -a_1 & 1 \\ -a_2 & 0 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{bmatrix} b_1 - a_1 b_0 \\ b_2 - a_2 b_0 \end{bmatrix} u$$
$$y = [1 \quad 0] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + [b_0] u$$

State-space model

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}$$

SISO:

$$x \in \mathbb{R}^n, u \in \mathbb{R}^1, y \in \mathbb{R}^1$$

MIMO:

$$x \in \mathbb{R}^n, u \in \mathbb{R}^m, y \in \mathbb{R}^p$$

Frequency-domain

$$sX(s) - x(0^-) = AX(s) + BU(s) \quad \Longrightarrow \quad X(s) = (sI - A)^{-1}BU(s) + (sI - A)^{-1}x(0^-)$$

Output

$$Y(s) = G(s)U(s) + F(s)x(0^-)$$

$$G(s) = C(sI - A)^{-1}B + D, \quad F(s) = C(sI - A)^{-1}$$

State-space model

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}$$

Transfer-function

$$G(s) = C(sI - A)^{-1}B + D$$

Proper

$$\lim_{|s| \rightarrow \infty} (sI - A)^{-1} = 0 \quad \implies \quad \lim_{|s| \rightarrow \infty} G(s) = D$$

Characteristic equation: roots of

$$|sI - A| = \det(sI - A) = 0,$$

that is eigenvalues of A , are the poles of $G(s)$

State-space model

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}$$

Matrix exponential

$$\mathcal{L}^{-1}\{(sI - A)^{-1}\} = e^{At}, \quad t \geq 0$$

Impulse response

$$g(t) = \mathcal{L}^{-1}\{G(s)\} = \mathcal{L}^{-1}\{C(sI - A)^{-1}B + D\} = Ce^{At}B + D\delta(t), \quad t \geq 0$$

State-space model

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}$$

Matrix exponential

$$\mathcal{L}^{-1}\{(sI - A)^{-1}\} = e^{At}, \quad t \geq 0$$

Response to initial condition

$$y(t) = \mathcal{L}^{-1}\{F(s)x(0^-)\} = \mathcal{L}^{-1}\{C(sI - A)^{-1}x(0^-)\} = Ce^{At}x(0^-), \quad t \geq 0$$

Asymptotic stability

$$\operatorname{Re}(\lambda_i(A)) < 0 \quad \Leftrightarrow \quad \lim_{t \rightarrow \infty} e^{At} = 0$$

We say A is **Hurwitz**

State-space model

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}$$

Matrix exponential

$$\mathcal{L}^{-1}\{(sI - A)^{-1}\} = e^{At}, \quad t \geq 0$$

Response to initial condition

$$y(t) = \mathcal{L}^{-1}\{F(s)x(0^-)\} = \mathcal{L}^{-1}\{C(sI - A)^{-1}x(0^-)\} = Ce^{At}x(0^-), \quad t \geq 0$$

Asymptotic stability

If A is Hurwitz then

$$\lim_{t \rightarrow \infty} x(t) = \lim_{t \rightarrow \infty} e^{At} x(0^-) = 0 \quad \implies \quad \lim_{t \rightarrow \infty} y(t) = \lim_{t \rightarrow \infty} Ce^{At} x(0^-) = 0$$

5.3 Minimal State-Space Realizations

State-space model

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ \beta \end{bmatrix}, \quad C = [\alpha \quad 1], \quad D = \gamma$$

order: $n = 2$

Transfer-function

$$G(s) = C(sI - A)^{-1}B + D = \frac{\alpha + \beta}{s} + \frac{\alpha\beta}{s^2} + \gamma$$

$\alpha\beta = 0, \alpha + \beta \neq 0 \implies$ transfer-function has order 1

$\alpha = \beta = 0 \implies$ transfer-function has order 0

State-space realization may not be *minimal*

Keywords: *controllability* and *observability*

State-space model

$$A = -\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad C = -\begin{bmatrix} 0 & 1/2 \\ 1 & 1/2 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

order: $n = 2$ is minimal

Transfer-function

$$G(s) = \begin{bmatrix} 1 & \frac{s}{s+1} \\ \frac{s-1}{s+1} & \frac{s}{s+1} \end{bmatrix}$$

“looks” like order 1; $s = 0$ is a zero but $s = 1$ is not!

MIMO systems can be much more complicated!

5.4 Nonlinear Systems and Linearization

State-space nonlinear model

$$\begin{aligned}\dot{x}(t) &= f(x(t), u(t)) \\ y(t) &= g(x(t), u(t))\end{aligned}$$

Taylor series

$$\begin{aligned}f(x, u) &\approx f(\bar{x}, \bar{u}) + A(x - \bar{x}) + B(u - \bar{u}) \\ g(x, u) &\approx g(\bar{x}, \bar{u}) + C(x - \bar{x}) + D(u - \bar{u})\end{aligned}$$

where

$$A = \left. \frac{\partial f}{\partial x} \right|_{x=\bar{x}, u=\bar{u}}, \quad B = \left. \frac{\partial f}{\partial u} \right|_{x=\bar{x}, u=\bar{u}}, \quad C = \left. \frac{\partial g}{\partial x} \right|_{x=\bar{x}, u=\bar{u}}, \quad D = \left. \frac{\partial g}{\partial u} \right|_{x=\bar{x}, u=\bar{u}}$$

State-space nonlinear model

$$\begin{aligned}\dot{x}(t) &= f(x(t), u(t)) \\ y(t) &= g(x(t), u(t))\end{aligned}$$

Equilibrium point

$$(\bar{x}, \bar{u}) : \quad f(\bar{x}, \bar{u}) = 0$$

Deviations

$$\tilde{x}(t) = x(t) - \bar{x}, \quad \tilde{u}(t) = u(t) - \bar{u}, \quad \tilde{y}(t) = y(t) - g(\bar{x}, \bar{u})$$

Taylor series

$$\begin{aligned}f(x, u) &\approx f(\bar{x}, \bar{u}) + A(x - \bar{x}) + B(u - \bar{u}) = A\tilde{x} + B\tilde{u} \\ g(x, u) &\approx g(\bar{x}, \bar{u}) + C(x - \bar{x}) + D(u - \bar{u}) = C\tilde{x} + D\tilde{u}\end{aligned}$$

State-space nonlinear model

$$\begin{aligned}\dot{x}(t) &= f(x(t), u(t)) \\ y(t) &= g(x(t), u(t))\end{aligned}$$

Equilibrium point

$$(\bar{x}, \bar{u}) : \quad f(\bar{x}, \bar{u}) = 0$$

Deviations

$$\tilde{x}(t) = x(t) - \bar{x}, \quad \tilde{u}(t) = u(t) - \bar{u}, \quad \tilde{y}(t) = y(t) - g(\bar{x}, \bar{u})$$

State-space linearized model

$$\begin{aligned}\dot{\tilde{x}}(t) &= A \tilde{x}(t) + B \tilde{u}(t) \\ \tilde{y}(t) &= C \tilde{x}(t) + D \tilde{u}(t)\end{aligned}$$

State-space nonlinear model

$$\dot{x}(t) = f(x(t), u(t))$$

$$y(t) = g(x(t), u(t))$$

State-space linearized model

$$\dot{\tilde{x}}(t) = A \tilde{x}(t) + B \tilde{u}(t)$$

$$\tilde{y}(t) = C \tilde{x}(t) + D \tilde{u}(t)$$

Lemma 5.1 (Lyapunov)

If A is Hurwitz then there exists $\epsilon > 0$ such that for any $x(0)$ within $\|x(0) - \bar{x}\| < \epsilon$ and $u(t) = \bar{u}$ then $\lim_{t \rightarrow \infty} x(t) \rightarrow \bar{x}$, that is $x(t)$ converges to $\bar{x}$

If A has at least one eigenvalue with positive real part then there exists solutions that diverge from $\bar{x}$

Lemma 5.1 (Lyapunov)

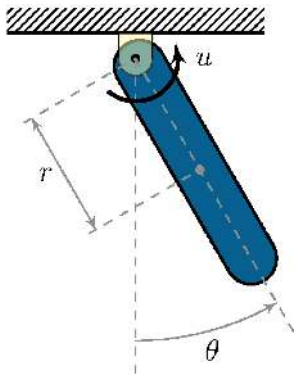
If A is Hurwitz then there exists $\epsilon > 0$ such that for any $x(0)$ within $\|x(0) - \bar{x}\| < \epsilon$ and $u(t) = \bar{u}$ then $\lim_{t \rightarrow \infty} x(t) \rightarrow \bar{x}$, that is $x(t)$ converges to $\bar{x}$

If A has at least one eigenvalue with positive real part then there exists solutions that diverge from $\bar{x}$

Key points:

- Stability of the linearized model implies stability of the nonlinear model “about” the equilibrium point
- No information about how large is the “region of attraction”
- Instability of the linearized model implies instability of the nonlinear model equilibrium point
- Inconclusive if eigenvalues are on the imaginary axis
- There is so much more to learn about nonlinear systems!

5.5 Simple Pendulum



Dynamic model

$$J_r \ddot{\theta} + b \dot{\theta} + m g r \sin \theta = u, \quad J_r = J + m r^2 > 0$$

Nonlinear state-space model

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} \dot{\theta} \\ \theta \end{pmatrix}, \quad f(x, u) = \begin{pmatrix} b_2 u - a_1 x_1 - a_2 \sin x_2 \\ x_1 \end{pmatrix}, \quad g(x, u) = x_2$$

Equilibrium points ($\bar{u} = 0$)

$$f(\bar{x}, \bar{u}) = \begin{pmatrix} -a_1 \bar{x}_1 - a_2 \sin \bar{x}_2 \\ \bar{x}_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies \bar{x}_1 = 0 \quad \text{and} \quad \bar{x}_2 = k\pi, \quad k \in \mathbb{Z}$$

Linearization

$$\frac{\partial f}{\partial x} = \begin{bmatrix} -a_1 & -a_2 \cos x_2 \\ 1 & 0 \end{bmatrix}, \quad \frac{\partial f}{\partial u} = \begin{bmatrix} b_2 \\ 0 \end{bmatrix}, \quad \frac{\partial g}{\partial x} = [0 \quad 1], \quad \frac{\partial g}{\partial u} = [0]$$

About

$$(\bar{x}, \bar{u}) = \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, 0 \right) : \quad A_0 = \begin{bmatrix} -a_1 & -a_2 \\ 1 & 0 \end{bmatrix}, \quad B_0 = \begin{bmatrix} b_2 \\ 0 \end{bmatrix}, \quad C_0 = [0 \quad 1], \quad D_0 = [0]$$

Transfer-function

$$G_0(s) = C_0(sI - A_0)^{-1}B_0 + D_0 = \frac{b_2}{s^2 + a_1 s + a_2}$$

Equilibrium points ($\bar{u} = 0$)

$$f(\bar{x}, \bar{u}) = \begin{pmatrix} -a_1 \bar{x}_1 - a_2 \sin \bar{x}_2 \\ \bar{x}_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies \bar{x}_1 = 0 \quad \text{and} \quad \bar{x}_2 = k\pi, \quad k \in \mathbb{Z}$$

Linearization

$$(\bar{x}, \bar{u}) = \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, 0 \right) : \quad A_0 = \begin{bmatrix} -a_1 & -a_2 \\ 1 & 0 \end{bmatrix}, \quad B_0 = \begin{bmatrix} b_2 \\ 0 \end{bmatrix}, \quad C_0 = [0 \quad 1], \quad D_0 = [0]$$

Transfer-function

$$G_0(s) = C_0(sI - A_0)^{-1}B_0 + D_0 = \frac{b_2}{s^2 + a_1 s + a_2}$$

If $a_1 > 0$ and $a_2 > 0$ then A_0 is Hurwitz, G_0 is asymptotically stable

Equilibrium points ($\bar{u} = 0$)

$$f(\bar{x}, \bar{u}) = \begin{pmatrix} -a_1 \bar{x}_1 - a_2 \sin \bar{x}_2 \\ \bar{x}_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies \bar{x}_1 = 0 \quad \text{and} \quad \bar{x}_2 = k\pi, \quad k \in \mathbb{Z}$$

Linearization

$$\frac{\partial f}{\partial x} = \begin{bmatrix} -a_1 & -a_2 \cos x_2 \\ 1 & 0 \end{bmatrix}, \quad \frac{\partial f}{\partial u} = \begin{bmatrix} b_2 \\ 0 \end{bmatrix}, \quad \frac{\partial g}{\partial x} = [0 \quad 1], \quad \frac{\partial g}{\partial u} = [0]$$

About

$$(\bar{x}, \bar{u}) = \left(\begin{pmatrix} 0 \\ \pi \end{pmatrix}, 0 \right) : \quad A_\pi = \begin{bmatrix} -a_1 & a_2 \\ 1 & 0 \end{bmatrix}, \quad B_\pi = \begin{bmatrix} b_2 \\ 0 \end{bmatrix}, \quad C_\pi = [0 \quad 1], \quad D_\pi = [0]$$

Transfer-function

$$G_\pi(s) = C_\pi(sI - A_\pi)^{-1}B_\pi + D_\pi = \frac{b_2}{s^2 + a_1s - a_2}$$

Equilibrium points ($\bar{u} = 0$)

$$f(\bar{x}, \bar{u}) = \begin{pmatrix} -a_1 \bar{x}_1 - a_2 \sin \bar{x}_2 \\ \bar{x}_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies \bar{x}_1 = 0 \quad \text{and} \quad \bar{x}_2 = k\pi, \quad k \in \mathbb{Z}$$

Linearization

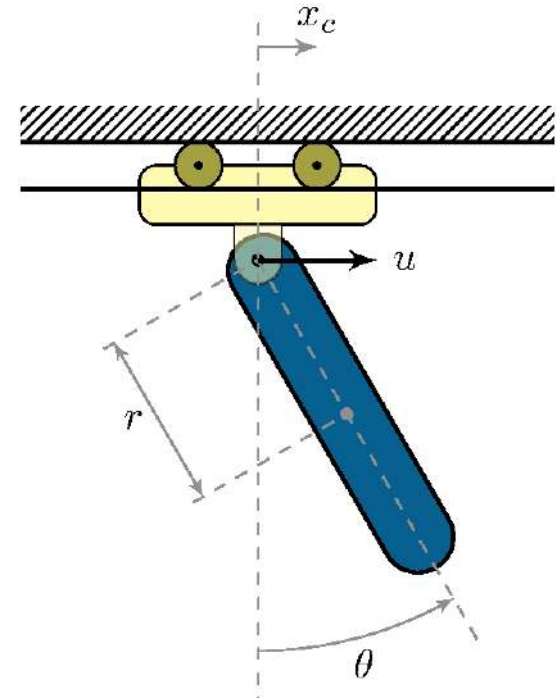
$$(\bar{x}, \bar{u}) = \left(\begin{pmatrix} 0 \\ \pi \end{pmatrix}, 0 \right) : \quad A_\pi = \begin{bmatrix} -a_1 & a_2 \\ 1 & 0 \end{bmatrix}, \quad B_\pi = \begin{bmatrix} b_2 \\ 0 \end{bmatrix}, \quad C_\pi = [0 \quad 1], \quad D_\pi = [0]$$

Transfer-function

$$G_\pi(s) = C_\pi(sI - A_\pi)^{-1}B_\pi + D_\pi = \frac{b_2}{s^2 + a_1s - a_2}$$

If $a_1 > 0$ and $a_2 > 0$ then A_π is not Hurwitz, G_π is unstable

5.6 Pendulum in a Cart



Dynamic model

$$\begin{aligned} (J_p + m_p r^2) \ddot{\theta} + m_p r \ddot{x}_c \cos \theta + b_p \dot{\theta} + m_p g r \sin \theta &= 0 \\ m_p r \ddot{\theta} \cos \theta + (m_p + m_c) \ddot{x}_c + b_c \dot{x}_c - m_p r \dot{\theta}^2 \sin \theta &= u \end{aligned}$$

Second-order vector differential equation

$$M(q) \ddot{q} + F(q, \dot{q}) = Gu, \quad q = \begin{pmatrix} \theta \\ x_c \end{pmatrix}$$

$$M(q) = \begin{bmatrix} J_p + m_p r^2 & m_p r \cos q_1 \\ m_p r \cos q_1 & m_p + m_c \end{bmatrix}, \quad F(q, \dot{q}) = \begin{pmatrix} b_p \dot{q}_1 + m_p g r \sin q_1 \\ b_c \dot{q}_2 - m_p r \dot{q}_1^2 \sin q_1 \end{pmatrix}, \quad G = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Nonlinear state-space model

$$x = \begin{pmatrix} q \\ \dot{q} \end{pmatrix}, \quad f(x, u) = f(q, \dot{q}, u) = \begin{pmatrix} \dot{q} \\ M^{-1}(q) [Gu - F(q, \dot{q})] \end{pmatrix}$$

Simplified nonlinear state-space model

$$x = \begin{pmatrix} \theta \\ \dot{\theta} \\ \dot{x}_c \end{pmatrix}, \quad f(x, u) = \begin{pmatrix} \dot{x}_1 \\ M^{-1}(x_1) [Gu - F(x)] \end{pmatrix}$$

Equilibrium points ($\bar{u} = 0$)

$$\bar{x}_1 = \bar{q}_1 = k\pi, \quad k \in \mathbb{Z}, \quad \bar{x}_2 = \bar{x}_3 = \bar{\dot{q}}_1 = \bar{\dot{q}}_2 = 0$$

Linearization about $\bar{x}_1 = 0$

$$A_0 = \frac{1}{J} \begin{bmatrix} 0 & J & 0 \\ -g m_r r & -b_p m_r / m_p & b_c r \\ g m_p r^2 & b_p r & -b_c J_r / m_p \end{bmatrix}, \quad B_0 = \frac{1}{J} \begin{bmatrix} 0 \\ -r \\ J_r / m_p \end{bmatrix}$$

$$J_r = J_p + m_p r^2 > 0, \quad m_r = m_p + m_c > 0, \quad J = J_r \frac{m_r}{m_p} - m_p r^2 > 0$$

Without damping ($b_c = b_p = 0$)

$$\text{Eigenvalues of } A_0 : \quad 0, \quad j\sqrt{g(m_p + m_c)r}, \quad -j\sqrt{g(m_p + m_c)r}$$

Oscillator

Equilibrium points ($\bar{u} = 0$)

$$\bar{x}_1 = \bar{q}_1 = k\pi, \quad k \in \mathbb{Z}, \quad \bar{x}_2 = \bar{x}_3 = \bar{\dot{q}}_1 = \bar{\dot{q}}_2 = 0$$

Linearization about $\bar{x}_1 = \pi$

$$A_\pi = \frac{1}{J} \begin{bmatrix} 0 & J & 0 \\ g m_r r & -b_p m_r / m_p & -b_c r \\ g m_p r^2 & -b_p r & -b_c J_r / m_p \end{bmatrix}, \quad B_\pi = \frac{1}{J} \begin{bmatrix} 0 \\ r \\ J_r / m_p \end{bmatrix}$$

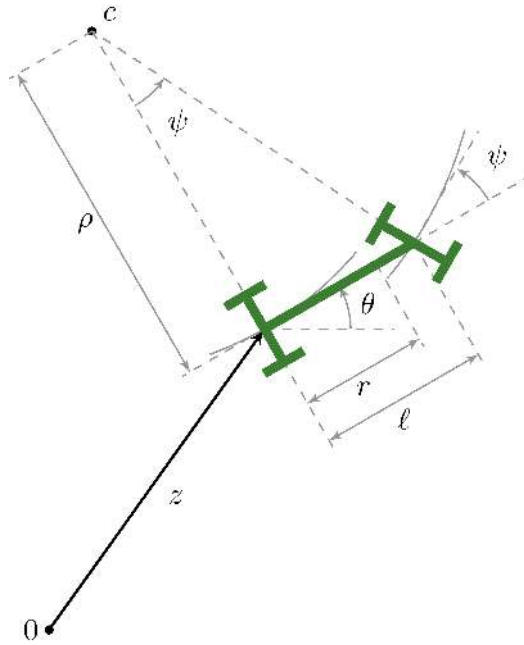
$$J_r = J_p + m_p r^2 > 0, \quad m_r = m_p + m_c > 0, \quad J = J_r \frac{m_r}{m_p} - m_p r^2 > 0$$

Without damping ($b_c = b_p = 0$)

$$\text{Eigenvalues of } A_\pi : \quad 0, \quad \sqrt{g(m_t + m_c)r}, \quad -\sqrt{g(m_t + m_c)r}$$

Unstable

5.7 Car Steering



www.toyhalloffame.org

Nonlinear state-space model (kinematic)

$$x = \begin{pmatrix} z_x \\ z_y \\ \theta \end{pmatrix}, \quad u = \tan \psi, \quad f(x, u) = \begin{pmatrix} v \cos \theta \\ v \sin \theta \\ uv/\ell \end{pmatrix}$$

Nonlinear state-space model

Kinematics

$$x = \begin{pmatrix} z_x \\ z_y \\ \theta \end{pmatrix}, \quad u = \tan \psi, \quad f(x, u) = \begin{pmatrix} v \cos \theta \\ v \sin \theta \\ uv/\ell \end{pmatrix}$$

Linearization

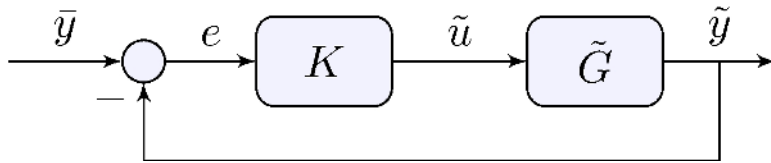
$$\frac{\partial f}{\partial x} = \begin{bmatrix} 0 & 0 & -v \sin \theta \\ 0 & 0 & v \cos \theta \\ 0 & 0 & 0 \end{bmatrix}, \quad \frac{\partial f}{\partial u} = \begin{bmatrix} 0 \\ 0 \\ v/\ell \end{bmatrix}$$

About moving trajectory (horizontal line)

$$\bar{x}(t) = \begin{pmatrix} z_x(t) \\ z_y(t) \\ \theta(t) \end{pmatrix} = \begin{pmatrix} vt \\ 0 \\ 0 \end{pmatrix}, \quad \bar{u}(t) = 0 : \quad A(t) = A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & v \\ 0 & 0 & 0 \end{bmatrix}, \quad B(t) = B = \begin{bmatrix} 0 \\ 0 \\ v/\ell \end{bmatrix}$$

5.8 Linear Control of Nonlinear Systems

Design



Linear controller K is designed based on the linearized model $\tilde{G}$

$$\tilde{u} = K(\bar{y} - \tilde{y}), \quad \tilde{u} = u - \bar{u}, \quad \tilde{y} = y - g(\bar{x}, \bar{u})$$

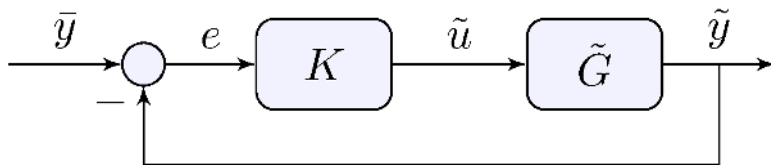
Implementation

Needs shifting

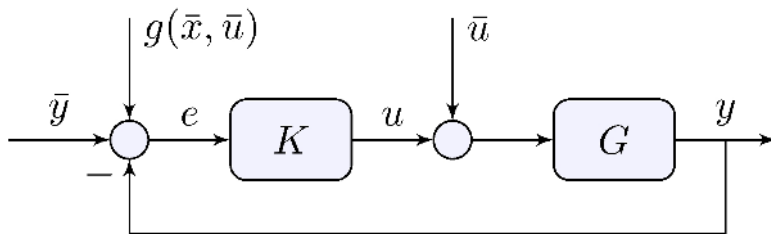
$$u = \bar{u} + K[\bar{y} + g(\bar{x}, \bar{u}) - y]$$

5.8 Linear Control of Nonlinear Systems

Design



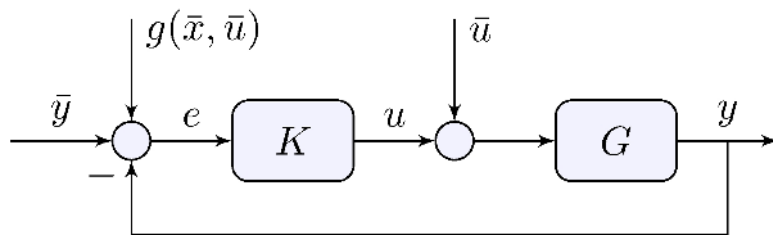
Implementation



Shifted set-point and control

$$u = \bar{u} + Ke, \quad e = \bar{y} + g(\bar{x}, \bar{u}) - y$$

Linear Control of Nonlinear Systems



Washout filter

Controller acts on transient but does not affect steady-state equilibrium ($K(0) = 0$)

Integral control shifts the equilibrium

Linearize about shifted equilibrium to check for stability

Learn more

- [Book website](#)